# Trajectory Design Employing Convex Optimization for Landing on Irregularly Shaped Asteroids

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**The problem under investigation is how to rapidly design a propellant optimal powered descent trajectory to land a spacecraft on an arbitrarily shaped asteroid, subject to realistic constraints on the trajectory and thrust magnitude, and a gravity model applicable to irregularly shaped asteroids. For a given time of flight, the problem is solved by finding the solutions to a sequence of convex optimization problems. A univariate optimization method is wrapped around the convex optimization problem to determine the optimal flight time that yields the overall lowest propellant usage. This solution methodology is robust and reliable. It does not require a user-supplied initial guess and no trial and error in selecting problem parameters. The strengths of this method are demonstrated by extensive numerical results by designing the propellant optimal trajectories for landing on the elongated binary asteroid Castalia.**

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## I. Introduction

MISSION proposals that land spacecraft on asteroids are becoming increasingly popular. However, in order to have a successful mission the spacecraft must reliably and softly land at the intended landing site with pinpoint precision. The problem under investigation is how to design a propellant optimal powered descent trajectory that can be quickly computed onboard the spacecraft, without interaction from ground control. An optimal trajectory designed immediately before the powered descent burn has many advantages. These advantages include the ability to use the actual vehicle state as the initial condition in the trajectory design and the ease of updating the target landing site. For long trajectories, the trajectory can be updated periodically by a redesign of the optimal trajectory based on current vehicle conditions to improve performance. One of the key drivers for being completely autonomous is the infrequent and delayed communication between the ground control and the vehicle. Challenges that arise from designing an asteroid powered descent trajectory include complicated nonlinear gravity fields associated with irregular shapes, small rotating bodies, and the low thrust of the vehicle. There are many factors that will not be well understood until the spacecraft reaches the asteroid; therefore, it is imperative to make the trajectory design algorithm adaptive to account for the updated information.

The propellant (fuel) optimal control problem in this work is to determine the optimal thrust vector (both magnitude and direction) to land the spacecraft at a specified location, in the presence of a highly nonlinear gravity field and subject to various mission and operational constraints. Based on the original nonlinear and nonconvex formulation for a fixed final flight time problem, this paper formulates a sequence of second-order cone programs, which are convex optimization problems. Included is a discussion on asteroid gravity models and their incorporation into the convex optimization formulation. Relaxation techniques, investigation on the equivalence of optimal solutions, and a successive solution method are all used to generate the final optimization problem that can be solved with a convex optimizer. Then, the investigation will focus on finding the optimal flight time, which results in minimum propellant usage over all flight times. Triaxial ellipsoidal asteroids (which have a simpler gravity model) are used to first investigate the trends in the optimal trajectories. The full methodology is demonstrated by finding optimal landing trajectories on the binary asteroid Castalia. Earlier work by the authors in [1] focused on the initial problem development and only the triaxial ellipsoidal asteroids.

There have been two successful spacecraft landings on asteroids and one on a comet. Near Earth Asteroid Rendezvous (NEAR) was the first successful asteroid landing when it landed on Eros. Four braking maneuvers were executed over a 50-min period to target a specified velocity decrease. The spacecraft impacted earlier than expected in the range of 1.5–1.8 m/s, slightly larger than the planned 1.3 m/s impact velocity [2]. Because of malfunctioning equipment, Hayabusa was painstakingly guided to a point near the landing site from ground control, even with a 20-min communication delay. Close to the surface, landing target markers were dropped to the surface and Hayabusa rendezvoused with the markers to successfully reach the surface [3]. Rosetta's lander Philae followed a ballistic trajectory down to the comet's surface [4].

There are a handful of proposed missions to asteroids whose landing algorithms and missions are still being finalized. Most rely on ballistic trajectories or ground designed trajectories and waypoints. The lander MASCOT relies on the primary spacecraft to drop down to 100 m altitude and then MASCOT free falls to the surface for the Marco Polo and Hayabusa-2 missions [5,6]. OSIRIS-REx has a checkpoint and a matchpoint at which maneuvers will be performed. These are targeted maneuvers, which are adjusted onboard to account for differences between the originally designed points and the actual spacecraft conditions at those points. The last maneuver at the matchpoint is 8 min before touchdown at 45 m altitude and slows the descent velocity [7].

Since the asteroid powered descent trajectory design is a relatively new topic of study, there are a number of proposed methods and ideas in development, all with advantages and disadvantages. The trajectory design and related guidance algorithms are both studied, as some methods forgo trajectory design and focus entirely on closed loop guidance efforts. One proposed approach to the rapid generation of the descent trajectory is a semi-analytical approach [8], which assumes that the acceleration profile is a cubic polynomial. Given the acceleration profile shape with known initial and final points, the problem is solved analytically except for three parameters: time of

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flight, initial thrust, and initial thrust angle. Combining these with path constraints yields a nonlinear program. A second approach for designing the descent trajectory focuses on extracting information from the optimal control theory problem formulation. An indirect method that propagates and solves the costates, boundary conditions, and differential equations is combined with a direct method that optimizes the original cost function. This combination decreases the computation time as compared with the purely direct method [9]. A time optimal asteroid landing problem is investigated in [10], where the thrust magnitude is assumed to always be at a single level. As such the problem is considerably simpler than a fuel optimal problem where multiple switchings of the optimal thrust magnitude between its upper and lower bounds are likely.

Variations on using sliding mode control techniques as a guidance algorithm for the descent trajectory are found in several places. One method uses multiple sliding surfaces to drive the spacecraft to the landing site in a finite amount of time [11]. Other proposals use a predesigned trajectory and apply variations on sliding mode control theory to counteract disturbances while tracking the trajectory [12,13]. One proposed method of predesigning the trajectory is to apply a homotopic and successive solution technique while solving the state and costate differential equations via a shootout method [12].

There are two previous studies that form the background to the current investigation. The first study looked in-depth at applying convex optimization to a powered descent trajectory on Mars with promising results [14,15]. This showed that the powered descent equations of motion can be relaxed and formed into a convex optimization problem and that the optimal solution of the relaxed problem is indeed a feasible solution to the original problem. This analysis used a constant gravity field, which is a poor approximation for an asteroid. In addition, the inclusion of thrust pointing constraints, while achieving lossless convexification, was examined in [16]. The second study applied a successive solution process to formulate a second-order cone program that designs rendezvous and proximity operations trajectories. This algorithm included a Newtonian gravity model. The equivalence of the solutions between the relaxed and the original problem was theoretically established [17]. Convex optimization had great success in the above studies and the focus of this analysis is the applicability to the asteroid powered descent problem.

Convex optimization is a class of optimization problems that includes many subclasses. It has been studied for the last hundred years; however, it has become popular in industry during the last 25 years with great success. Applications of convex optimization in industry include estimation and signal processing, modeling statistics, finance, and automatic control systems. A convex optimization problem can be solved reliably and efficiently. If the problem is feasible, then the global optimal can be found in a finite number of steps. If the problem is infeasible, then no solution is possible. Convex optimization includes the subclasses of linear program, second-order cone program (SOCP), quadratic program, and least squares. There are publicly and commercially available solvers for convex optimization problems along with numerous published methods [18]. If an optimization problem can be formulated as a convex optimization problem, it can be easily solved. The challenge is to turn a nonlinear optimization problem into a convex optimization problem.

## II. Propellant Optimal Powered Descent Formulation

### A. Asteroid Gravitational Models

Asteroids come in a wide variety of sizes and shapes, which are rarely spherical. Therefore, the standard gravity models of constant gravity and Newtonian gravity that are good approximations in powered descent algorithms on Earth, Moon, and Mars are poor models for an asteroid powered descent mission. This insufficiency was shown through analysis that designed the trajectory with these models and then simulated the vehicle flight using a higher fidelity gravity model. The vehicle did not reach the asteroid surface in the majority of cases (see [19]). The best gravity model for an asteroid is still a hotly researched topic with a wide range of models under study and new models being published on a regular basis.

For a homogenous triaxial ellipsoid, a good starting model is the $2 \times 2$ spherical harmonics model (maximum degree and order of 2). This is a readily available model (see [20]). The spherical harmonics gravitational model ($\nabla U(\boldsymbol{r})$) for all degrees and order is listed in Eqs. (1–3). The equations of motion for the propellant optimal problem are in a Cartesian coordinate system, whereas the spherical harmonics model is in the spherical coordinate system. Therefore, the partials of the gravitational potential with respect to the spherical coordinates, Eq. (2), and the partials of position vector in the spherical coordinate system ($r$, $\delta$, $\lambda$) with respect to the position vector in the Cartesian coordinate system ($r_x$, $r_y$, $r_z$), Eq. (3), are required.

$$\begin{aligned}
\frac{\partial U}{\partial r_x} &= \frac{\partial U}{\partial r}\frac{\partial r}{\partial r_x} + \frac{\partial U}{\partial \delta}\frac{\partial \delta}{\partial r_x} + \frac{\partial U}{\partial \lambda}\frac{\partial \lambda}{\partial r_x} \\
\frac{\partial U}{\partial r_y} &= \frac{\partial U}{\partial r}\frac{\partial r}{\partial r_y} + \frac{\partial U}{\partial \delta}\frac{\partial \delta}{\partial r_y} + \frac{\partial U}{\partial \lambda}\frac{\partial \lambda}{\partial r_y} \\
\frac{\partial U}{\partial r_z} &= \frac{\partial U}{\partial r}\frac{\partial r}{\partial r_z} + \frac{\partial U}{\partial \delta}\frac{\partial \delta}{\partial r_z} + \frac{\partial U}{\partial \lambda}\frac{\partial \lambda}{\partial r_z}
\end{aligned} \tag{1}$$

$$\begin{aligned}
\frac{\partial U}{\partial r} &= \sum_{l=0}^{N}\sum_{m=0}^{l} -(l+1)\frac{\mu}{r^2}\left(\frac{r_0}{r}\right)^l P_{l,m}[\sin\delta]\{C_{l,m}\cos(m\lambda) + S_{l,m}\sin(m\lambda)\} \\
\frac{\partial U}{\partial \delta} &= \sum_{l=0}^{N}\sum_{m=0}^{l} \frac{\mu}{r}\left(\frac{r_0}{r}\right)^l \{C_{l,m}\cos(m\lambda) + S_{l,m}\sin(m\lambda)\}\frac{\partial P_{l,m}[\sin\delta]}{\partial \delta} \\
\frac{\partial U}{\partial \lambda} &= \sum_{l=0}^{N}\sum_{m=0}^{l} \frac{\mu}{r}\left(\frac{r_0}{r}\right)^l m P_{l,m}[\sin\delta]\{-C_{l,m}\sin(m\lambda) + S_{l,m}\cos(m\lambda)\}
\end{aligned} \tag{2}$$

$$\begin{aligned}
&\frac{\partial r}{\partial r_x} = \frac{r_x}{r}, \quad \frac{\partial r}{\partial r_y} = \frac{r_y}{r}, \quad \frac{\partial r}{\partial r_z} = \frac{r_z}{r} \\
&\frac{\partial \delta}{\partial r_x} = \frac{-r_x r_z}{r^2\sqrt{r_x^2 + r_y^2}}, \quad \frac{\partial \delta}{\partial r_y} = \frac{-r_y r_z}{r^2\sqrt{r_x^2 + r_y^2}}, \\
&\frac{\partial \delta}{\partial r_z} = \frac{1}{\sqrt{r_x^2 + r_y^2}}\left(1 - \frac{r_z^2}{r^2}\right) \\
&\frac{\partial \lambda}{\partial r_x} = \frac{-r_y}{r_x^2 + r_y^2}, \quad \frac{\partial \lambda}{\partial r_y} = \frac{r_x}{r_x^2 + r_y^2}, \quad \frac{\partial \lambda}{\partial r_z} = 0
\end{aligned} \tag{3}$$

The gravitational model depends on associated Legendre functions ($P_{l,m}[\sin\delta]$) and the gravitational coefficients ($C_{l,m}$, $S_{l,m}$). The gravitational coefficients are determined for each asteroid. For a $2 \times 2$ model, $N = 2$ in Eq. (2).

The homogenous triaxial ellipsoidal assumptions of constant density and symmetry around the axes causes the only nonzero coefficients to be the $C_{l,m}$ coefficients that have an even $l$ and $m$. Included is the assumption that the coordinate system origin is located at the asteroid's center of mass and the axes are aligned with the semi-major axes. The homogeneous triaxial ellipsoidal model was used in the algorithm proof-of-concept test before transitioning to even higher-fidelity models. The homogeneous, ellipsoidal shape, and axis alignment and placement assumptions can be removed by including all the spherical harmonic gravitational coefficients in the expansion and calculating the coefficients based on high-fidelity shape models.

Unfortunately, there is an underlying implicit assumption in the spherical harmonics gravity model. It assumes that the mathematical series representing the gravitational field converges to the actual gravitational field. If the point where the field is evaluated is outside the maximum radius of mass distribution for the asteroid, this assumption is valid, but when a point is evaluated inside this maximum radius the mathematical series diverges from the true gravitational field. The sphere circumscribing the asteroid mass is called the Brillouin sphere. An example of the Brillouin sphere with an irregularly shaped asteroid is shown in Fig. 1. The spherical harmonics potential model is valid outside of the Brillouin sphere.

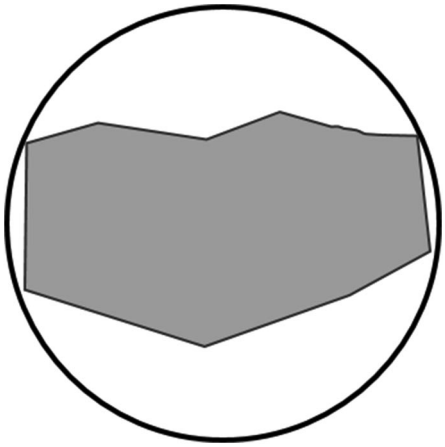

**Fig. 1 Brillouin sphere surrounding an irregularly shaped asteroid.**

Inside the Brillouin sphere a different model must be used. The more irregularly shaped the asteroid, or the deeper the valleys near the landing site, the worse the divergence [21,22].

A wide range of gravitational models pertaining to asteroids have been published over the last few decades, with new ones appearing every couple of years. Some of the more widely recognized models are the homogeneous polyhedron [23,24], mascon model [22,24], and several interior models [21,25].

The polyhedron method is considered to be the most accurate model and is used as the benchmark for comparing new gravity models to determine their accuracy. However, this is a very computationally expensive proposition for an entire trajectory, as each trajectory point would require tens of thousands of calculations. This model is best suited on the ground with fast computers and adequate time to design the trajectory, but it is not suitable for a flight computer and the need for rapid trajectory design. Therefore, it is not a candidate for the autonomous powered descent trajectory design. Another feature of the polyhedron method is that it can be used to determine the coefficients for the spherical harmonics expansion model and the interior gravity models [23,24].

The best choice for a gravity model that balances computational complexity and accuracy is to use a full $4 \times 4$ spherical harmonics model [$N = 4$ in Eq (2)] outside the Brillouin sphere and transition to the interior spherical Bessel gravity model at the sphere and inside the Brillouin sphere. The interior spherical Bessel gravity model is valid inside the Brillouin sphere down to the asteroid surface. A brief overview of the interior spherical Bessel gravity model is found in Eqs. (4–7). For definition of the terms, for example, $\bar{\mathscr{F}}_1(n,m)$, $\bar{\mathscr{F}}_2(n,m)$, $\bar{\mathscr{F}}_3(n,m)$, please see the model published in its original form in [25].

The gravitational acceleration is calculated in Eq. (4). It is a summation series over the $l$—power, $n$—degree, and $m$—order. A bar over a variable indicates that the associated value has been normalized.

$$\begin{aligned}
\frac{\partial U}{\partial r_x} &= \frac{\mu}{R_b}\sum_{l=0}^{\infty}\sum_{n=0}^{\infty}\sum_{m=0}^{n} Re\left[\frac{\partial}{\partial x}(\bar{\beta}_{n,m}(\alpha_{l,n}))\right]\bar{A}_{l,n,m} \\
&\quad + Im\left[\frac{\partial}{\partial x}(\bar{\beta}_{n,m}(\alpha_{l,n}))\right]\bar{B}_{l,n,m} \\
\frac{\partial U}{\partial r_y} &= \frac{\mu}{R_b}\sum_{l=0}^{\infty}\sum_{n=0}^{\infty}\sum_{m=0}^{n} Re\left[\frac{\partial}{\partial y}(\bar{\beta}_{n,m}(\alpha_{l,n}))\right]\bar{A}_{l,n,m} \\
&\quad + Im\left[\frac{\partial}{\partial y}(\bar{\beta}_{n,m}(\alpha_{l,n}))\right]\bar{B}_{l,n,m} \\
\frac{\partial U}{\partial r_z} &= \frac{\mu}{R_b}\sum_{l=0}^{\infty}\sum_{n=0}^{\infty}\sum_{m=0}^{n} Re\left[\frac{\partial}{\partial z}(\bar{\beta}_{n,m}(\alpha_{l,n}))\right]\bar{A}_{l,n,m} \\
&\quad + Im\left[\frac{\partial}{\partial z}(\bar{\beta}_{n,m}(\alpha_{l,n}))\right]\bar{B}_{l,n,m}
\end{aligned} \tag{4}$$

The normalized gravitational coefficients $\bar{A}_{l,n,m}$ and $\bar{B}_{l,m,n}$ are asteroid specific, similar to the $C_{l,m}$ and $S_{l,m}$ in the spherical harmonics model. As the acceleration vector must be real, the imaginary basis vector is parsed into its real and imaginary portions, represented by $Re$ and $Im$. The basis function in Eq. (5) relies on associated Legendre functions ($P_{n,m}$) and spherical Bessel functions ($j_n$). The eigenvalue of the spherical Bessel function is represented by $\alpha_{l,n}$.

$$\bar{\beta}_{n,m}(\alpha_{l,n}) = j_n\left(\frac{\alpha_{l,n} r}{R_b}\right)\bar{H}_{n,m} \tag{5}$$

$$\bar{H}_{n,m} = \begin{cases} N_{n,m}P_{n,m}[\sin(\phi)]e^{im\lambda} & n \geq m \geq 0 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

The partial derivatives of the basis function with respect to the position vector in the Cartesian coordinate system are located in Eq. (7).

$$\frac{\partial}{\partial x}\left(\bar{\beta}_{n,m}(\alpha_{l,n})\right) = \begin{cases} -\dfrac{\alpha_{l,n} r_x}{R_b r} j_{n+1}\left[\dfrac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n,0} & \\ -2\bar{\mathscr{F}}_1(n,m)\dfrac{1}{r} j_n\left[\dfrac{\alpha_{l,n} r}{R_b}\right]N_{n-1,1}P_{n-1,1}[\sin(\phi)]\cos\lambda & m = 0 \\ -\dfrac{\alpha_{l,n} r_x}{R_b r} j_{n+1}\left[\dfrac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n,m} - \bar{\mathscr{F}}_1(n,m)\dfrac{1}{r} j_n\left[\dfrac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n-1,m+1} + & \\ \bar{\mathscr{F}}_2(n,m)\dfrac{1}{r} j_n\left[\dfrac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n-1,m-1} & m > 0 \end{cases}$$

$$\frac{\partial}{\partial y}\left(\bar{\beta}_{n,m}(\alpha_{l,n})\right) = \begin{cases} -\dfrac{\alpha_{l,n} r_y}{R_b r} j_{n+1}\left[\dfrac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n,0} & \\ -2\bar{\mathscr{F}}_1(n,m)\dfrac{1}{r} j_n\left[\dfrac{\alpha_{l,n} r}{R_b}\right]N_{n-1,1}P_{n-1,1}[\sin(\phi)]\sin\lambda & m = 0 \\ -\dfrac{\alpha_{l,n} r_y}{R_b r} j_{n+1}\left[\dfrac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n,m} + \bar{\mathscr{F}}_1(n,m)\dfrac{1}{r} j_n\left[\dfrac{\alpha_{l,n} r}{R_b}\right]i\bar{H}_{n-1,m+1} + & \\ \bar{\mathscr{F}}_2(n,m)\dfrac{1}{r} j_n\left[\dfrac{\alpha_{l,n} r}{R_b}\right]i\bar{H}_{n-1,m-1} & m > 0 \end{cases}$$

$$\frac{\partial}{\partial z}\left(\bar{\beta}_{n,m}(\alpha_{l,n})\right) = -\frac{\alpha_{l,n} r_z}{R_b r} j_{n+1}\left[\frac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n,m} + \bar{\mathscr{F}}_3(n,m)\frac{1}{r} j_n\left[\frac{\alpha_{l,n} r}{R_b}\right]\bar{H}_{n-1,m} \tag{7}$$

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The summation series truncation used for the analysis is to a maximum power of 2 and degree and order of 5, based on the recommendation of the model's developers. Coefficients are developed using the available asteroid information, such as a polyhedron model. The fidelity of the models used to develop the coefficients will determine the maximum available degree, order, and power, along with the quantity of terms necessary to yield a specific accuracy.

The similarity between the spherical harmonics and the spherical Bessel models is one reason why this model was chosen as the gravity model inside the Brillouin sphere. When the coefficients for both models are derived together from the corresponding polyhedron model, there will be a seamless transition between the two models. However, in the case where the coefficients are not derived from the same source, the powered descent algorithm works fine with a small discontinuity in gravitational acceleration at the Brillouin sphere.

The spherical Bessel model is a very accurate model as compared with other gravity models, bar the polyhedron model. The error for Bennu (near spherical) was less than 5%, whereas Castalia (elongated binary) was less than a 10% error, when compared with the polyhedron model. Since the interior spherical Bessel model is significantly less computationally intensive than the polyhedron, it makes a good candidate for onboard the spacecraft, given that it can achieve similar accuracy to the polyhedron model.

The $2 \times 2$ and $4 \times 4$ combined with the Bessel gravity models are functions of the spacecraft position vector and the coefficients associated with the asteroid. As seen above, the position components are numerous and highly nonlinear. The gravity model will generically be referred to as $\nabla U(\boldsymbol{r})$ in the optimization problem formulation. One interesting item to note is that all of the gravity models can be separated into a dominant term and remaining higher-order terms. For the $2 \times 2$ and $4 \times 4$ the dominant term is the degree and order of 0 term, which simplifies to the standard Newtonian gravity model, $-(\mu/r^3)\boldsymbol{r}$. The dominant term for the Bessel model is the summation over the powers with degree and order 0, which contains three terms for the three powers.

### B. Nonlinear Optimization Problem

The propellant optimal soft landing problem is a two-point boundary value problem, as the initial vehicle state is known along with the targeted landing site. Minimizing propellant usage for a fixed flight time ($t_f$) is the optimization portion. This is achieved by maximizing the mass of the vehicle when it touches down [Eq. (8)], thus minimizing propellant consumption. The optimization problem is stated in Eqs. (8–16), referred to as problem P1.

$$\min -m(t_f) \tag{8}$$

subject to

$$\dot{\boldsymbol{r}} = \boldsymbol{v} \tag{9}$$

$$\dot{\boldsymbol{v}} = \frac{\boldsymbol{T}}{m} - 2\boldsymbol{\omega} \times \boldsymbol{v} - \dot{\boldsymbol{\omega}} \times \boldsymbol{r} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) + \nabla U(\boldsymbol{r}) \tag{10}$$

$$\dot{m} = -\frac{1}{v_{\mathrm{ex}}} \|\boldsymbol{T}\| \tag{11}$$

$$T_{\min} \leq \|\boldsymbol{T}\| \leq T_{\max} \tag{12}$$

$$\|\boldsymbol{r} - \boldsymbol{r}_f\| \cos\theta - (\boldsymbol{r} - \boldsymbol{r}_f)^T \hat{n} \leq 0 \tag{13}$$

$$m \geq m_{\mathrm{dry}} \tag{14}$$

$$\boldsymbol{r}(0) = \boldsymbol{r}_0, \qquad \boldsymbol{v}(0) = \boldsymbol{v}_0, \qquad m(0) = m_{\mathrm{wet}} \tag{15}$$

$$\boldsymbol{r}(t_f) = \boldsymbol{r}_f, \qquad \boldsymbol{v}(t_f) = \boldsymbol{v}_f, \qquad t_f \text{ given} \tag{16}$$

The equations of motion, Eqs. (9) and (10), are expressed in an asteroid fixed (rotating) Cartesian coordinate system with the origin at the center of the mass of the asteroid. The state vector consists of the vehicle position vector ($\boldsymbol{r}$), the vehicle's asteroid relative velocity vector ($\boldsymbol{v}$), and the vehicle mass ($m$). The vehicle's initial state is represented by $\boldsymbol{r}_0$, $\boldsymbol{v}_0$, and $m_{\mathrm{wet}}$, whereas the final vehicle state (the landing site) is represented by $\boldsymbol{r}_f$ and $\boldsymbol{v}_f$. As the asteroid is rotating, landing sites not located on the asteroid rotation axis ($\boldsymbol{\omega}$) have a nonzero velocity due to the asteroid's rotation. Once the engine has been turned on it is throttleable between a minimum ($T_{\min}$) and maximum ($T_{\max}$) thrust and cannot be turned off until the descent burn is complete, Eq. (12). The right-hand inequality constraint in Eq. (12) is convex and a valid inequality for a convex optimization problem, whereas the left inequality is not. Along with the vehicle's thrust range, the vehicle exit velocity ($v_{\mathrm{ex}}$) is required for the mass flow rate calculation. The constraint in Eq. (13) is called a glide slope constraint. Initially, the angle $\theta$ is set at 90.0 deg to prevent the spacecraft from going below the surface of the asteroid. The unit vector normal to the landing site is $\hat{n}$. The final constraint, Eq. (14), is to ensure that the trajectory does not use more than the available propellant, as $m_{\mathrm{dry}}$ is the mass of the vehicle when all propellant has been consumed.

Problem P1 is nonconvex. The nonconvex terms are the norm of the thrust vector, dividing by the mass, and the gravitational acceleration. Therefore, these terms will be handled with a variety of techniques to create a convex optimization problem.

## III. Lossless Convexification

Initially, two techniques are applied to problem P1 to turn the problem into a convex optimization problem. The first is a relaxation of the problem by introducing a slack variable and the second technique is a change of variables. Ultimately, the goal is to achieve lossless convexification, where the solution to the convex problem is indeed a minimal point of the original problem.

A slack variable, $T_m$, representing the thrust magnitude, is chosen to be the fourth control variable to supplement $\boldsymbol{T}$ [14,16]. In the optimization problem this variable is separate from the thrust vector and only related through a new second-order cone constraint, Eq. (17).

$$\|\boldsymbol{T}\| \leq T_m \tag{17}$$

The optimization problem is now formulated as problem P2, located in Eqs. (18–27). The boxes highlight the differences between problems P1 and P2.

$$\min -m(t_f) \tag{18}$$

subject to

$$\dot{\boldsymbol{r}} = \boldsymbol{v} \tag{19}$$

$$\dot{\boldsymbol{v}} = \frac{\boldsymbol{T}}{m} - 2\boldsymbol{\omega} \times \boldsymbol{v} - \dot{\boldsymbol{\omega}} \times \boldsymbol{r} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) + \nabla U(\boldsymbol{r}) \tag{20}$$

$$\dot{m} = -\frac{1}{v_{\mathrm{ex}}} \boxed{T_m} \tag{21}$$

$$\boxed{\left\|\boldsymbol{T}\right\| \leq T_m} \tag{22}$$

$$T_{\min} \leq \boxed{T_m} \leq T_{\max} \tag{23}$$

$$\|\boldsymbol{r} - \boldsymbol{r}_f\| \cos\theta - (\boldsymbol{r} - \boldsymbol{r}_f)^T \hat{n} \leq 0 \tag{24}$$

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$$m \geq m_{\text{dry}} \tag{25}$$

$$\boldsymbol{r}(0) = \boldsymbol{r}_0, \qquad \boldsymbol{v}(0) = \boldsymbol{v}_0, \qquad m(0) = m_{\text{wet}} \tag{26}$$

$$\boldsymbol{r}(t_f) = \boldsymbol{r}_f, \qquad \boldsymbol{v}(t_f) = \boldsymbol{v}_f, \qquad t_f \text{ given} \tag{27}$$

*Proposition:* Problems P1 and P2 are identical when the inequality in Eq. (22) is active, that is, $\|\boldsymbol{T}\| = T_m$. Moreover, the solution to problem P2 is the same as the solution to (minimum of) problem P1.

*Proof:* For brevity of the proof, the glide slope constraint Eq. (24) is removed. The first part of the proposition is straightforward. It is immediately clear that when the constraint in Eq. (25) is active, problem P2 reduces to problem P1. To prove the second part of the proposition, apply the minimum principle from optimal control theory and form the Hamiltonian, Eq. (28).

$$H = \boldsymbol{p}_r^T \boldsymbol{v} + \boldsymbol{p}_v^T \left( \frac{1}{m}\boldsymbol{T} - 2W_1\boldsymbol{v} - W_2\boldsymbol{r} - W_3\boldsymbol{r} + \nabla U(\boldsymbol{r}) \right) - p_m \frac{1}{v_{\text{ex}}} T_m$$

$$W_1 = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad W_2 = \begin{bmatrix} 0 & -\dot{\omega}_z & \dot{\omega}_y \\ \dot{\omega}_z & 0 & -\dot{\omega}_x \\ -\dot{\omega}_y & \dot{\omega}_x & 0 \end{bmatrix}$$

$$W_3 = \begin{bmatrix} -\omega_y^2 - \omega_z^2 & \omega_x\omega_y & \omega_x\omega_z \\ \omega_x\omega_y & -\omega_x^2 - \omega_z^2 & \omega_y\omega_z \\ \omega_x\omega_z & \omega_y\omega_z & -\omega_x^2 - \omega_y^2 \end{bmatrix} \tag{28}$$

The states are $\boldsymbol{r}$, $\boldsymbol{v}$, $m$, and the corresponding costates are $\boldsymbol{p}_r$, $\boldsymbol{p}_v$, $p_m$. The control vector $\mathscr{C}$ consists of the thrust vector and slack variable, along with the constraints on the control vector as described in Eq. (29).

$$\mathscr{C} = \{(\boldsymbol{T}, T_m) | \|\boldsymbol{T}\| \leq T_m, T_{\min} \leq T_m, T_m \leq T_{\max}\} \tag{29}$$

From Optimal Control Theory, the differential equations for the costates are listed in Eqs. (30–32).

$$\dot{\boldsymbol{p}}_r = W_2^T \boldsymbol{p}_v + W_3^T \boldsymbol{p}_v - \boldsymbol{p}_v^T \nabla^2 U(\boldsymbol{r}) \tag{30}$$

$$\dot{\boldsymbol{p}}_v = -\boldsymbol{p}_r + 2W_1^T \boldsymbol{p}_v \tag{31}$$

$$\dot{p}_m = \boldsymbol{p}_v^T \boldsymbol{T} \left( \frac{1}{m^2} \right) \tag{32}$$

The problem is a fixed final time problem. Applying the end point constraints and the transversality conditions yields one useful equation, Eq. (33).

$$p_m(t_f) = -1 \tag{33}$$

From Minimum Principle, the optimal solution is determined by minimizing the Hamiltonian with respect to the control vector $\mathscr{C}$ subject to the constraints in Eq. (29). The Hamiltonian is linear in terms of the control vector, and therefore the minimum will be on the boundary of the admissible control set. This can be turned into a constrained optimization problem, which minimizes the Hamiltonian by applying the Kasrush–Kuhn–Tucker (KKT) conditions. First, the terms in the Hamiltonian expression that are not dependent on the control vector can be removed from the optimization problem, as they have no effect on the solution. These terms are $\boldsymbol{p}_r^T \boldsymbol{v} + \boldsymbol{p}_v^T(-2W_1\boldsymbol{v} - W_2\boldsymbol{r} - W_3\boldsymbol{r} + \nabla U(\boldsymbol{r}))$. It is interesting to note that as long as the gravitational potential remains solely a function of spacecraft position, it does not influence the minimization of the Hamiltonian. Therefore, both the $2 \times 2$ and the $4 \times 4$ combined with spherical Bessel models are included in the proof.

The point-wise minimization problem with respect to $\boldsymbol{T}$, $T_m$ is stated in Eq. (34).

$$\begin{aligned} &\min \ \frac{1}{m}\boldsymbol{p}_v^T \boldsymbol{T} - \frac{1}{v_{\text{ex}}} p_m T_m \\ &\text{s.t. } \|\boldsymbol{T}\| \leq T_m, \quad T_{\min} \leq T_m, \quad T_m \leq T_{\max} \end{aligned} \tag{34}$$

The Lagrangian for the constrained optimization problem is listed in Eq. (35).

$$\begin{aligned} \mathscr{L} = &\frac{1}{m}\boldsymbol{p}_v^T \boldsymbol{T} - \frac{1}{v_{\text{ex}}} p_m T_m + \lambda_1(\|\boldsymbol{T}\| - T_m) + \lambda_2(T_{\min} - T_m) \\ &+ \lambda_3(T_m - T_{\max}) \end{aligned} \tag{35}$$

The variables $\lambda_1$, $\lambda_2$, $\lambda_3$ are the Lagrange multipliers associated with the inequality constraints on $\boldsymbol{T}$, $T_m$. From the KKT conditions, a constraint is active when $\lambda_i \geq 0$, whereas a constraint can only be inactive when $\lambda_i = 0$. Applying the KKT conditions yields Eqs. (36) and (37).

$$\frac{\partial \mathscr{L}}{\partial \boldsymbol{T}} = \frac{1}{m}\boldsymbol{p}_v + \lambda_1 \frac{\boldsymbol{T}}{\|\boldsymbol{T}\|} = 0 \tag{36}$$

$$\frac{\partial \mathscr{L}}{\partial T_m} = -\frac{1}{v_{\text{ex}}} p_m - \lambda_1 - \lambda_2 + \lambda_3 = 0 \tag{37}$$

Now examine the problem as two specific cases: when $\boldsymbol{p}_v$ is nonzero almost everywhere (a.e.) in $[0, t_f]$ and when it is zero in a finite time interval. Suppose that $\boldsymbol{p}_v \neq 0$ for any finite time interval. From Eq. (36), when $\boldsymbol{T} \neq 0$, then $\lambda_1$ must be nonzero ($\lambda_1 \neq 0$); otherwise, Eq. (36) necessitates $\boldsymbol{p}_v = 0$ because vehicle mass is nonzero, thus contradicting the assumption that $\boldsymbol{p}_v \neq 0$. When $\boldsymbol{T} = 0$, $\boldsymbol{T}/\|\boldsymbol{T}\|$ is a finite vector. The same argument then applies to ascertain that $\lambda_1 \neq 0$. Therefore, the corresponding constraint must be active, $\|\boldsymbol{T}\| = T_m$. Consequently, the solution to problem P2 is also the solution to problem P1.

Next, look at the case when $\boldsymbol{p}_v = 0$ for a finite time period. If $\boldsymbol{p}_v = 0$ in a finite time interval, its derivative must also be zero simultaneously. From the costate equations in Eqs. (30–32), this implies that $\boldsymbol{p}_r = 0$ and $p_m$ is a constant. Because the costate equations are a homogeneous linear system in $\boldsymbol{p}_r$ and $\boldsymbol{p}_v$, it follows that $\boldsymbol{p}_r = \boldsymbol{p}_v = 0$ in the entire interval $[0, t_f]$, and $p_m$ is constant [26]. From the transversality condition in Eq. (33), $p_m(t) = -1$. The Hamiltonian reduces to Eq. (38).

$$H = \frac{1}{v_{\text{ex}}} T_m \tag{38}$$

The minimum of the Hamiltonian with respect to $\mathscr{C}$ is the smallest admissible $T_m$, which is $T_{\min} > 0$, and therefore $T_m = T_{\min}$. Thus the Hamiltonian becomes Eq. (39).

$$H = \frac{1}{v_{\text{ex}}} T_{\min} \tag{39}$$

If $\boldsymbol{p}_v = 0$ in a finite interval, the constancy condition of the Hamiltonian due to the autonomous system dynamics should always hold, regardless of the initial conditions, landing site, or the specified final time, $t_f$. In particular, if the value of $t_f$ happens to be equal to $t_f^*$—the optimal flight time in a free time problem, which is the same as problem P2 except that the final time is free—the Hamiltonian is zero along the optimal trajectory from Minimum Principle. The nonzero constancy condition in Eq. (39) then constitutes a contradiction in this case. Therefore, the condition that $\boldsymbol{p}_v = 0$ in a finite interval cannot happen. Since this case is invalid, the case of $\boldsymbol{p}_v \neq 0$ a.e. on $[0, t_f]$ is the only possibility, which in turn means that $\|\boldsymbol{T}\| = T_m$ in the solution to problem P2. This solution is also the solution to problem P1 [27].

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The next step after the slack variable introduction is to employ two different types of variable changes. The first changes all thrust variables into acceleration due to thrust through the conversion seen in Eq. (40). The second introduces a new mass variable defined in Eq. (41).

$$\boldsymbol{a}_t = \frac{\boldsymbol{T}}{m}, \qquad a_{\text{tm}} = \frac{T_m}{m} \tag{40}$$

$$q = \ln(m), \qquad \dot{q} = -\frac{a_{\text{tm}}}{v_{\text{ex}}} \tag{41}$$

The cost function will find the maximum of this new mass variable, as the maximum mass at the landing site corresponds to the maximum of $q$, due to the properties of the natural log function.

With these variable changes Eq. (23) becomes two nonlinear inequalities. As in [14], the Taylor's series expansion will be applied to approximate the nonlinear nonconvex constraints by a second-order cone constraint and a linear constraint, which are both convex. The new constraints are in Eq. (42), where the left side is a second-order Taylor series and the right side a first-order Taylor series. Spot checking shows the deviation $(q - q_o)$ is on the order of a few grams, which ensures the validity of the Taylor series approximations.

$$T_{\min} e^{-q_o}[1-(q-q_o)+0.5(q-q_o)^2] \le a_{\text{tm}} \le T_{\max} e^{-q_o}[1-(q-q_o)] \tag{42}$$

The point about which the expansion occurs $(q_o)$ is calculated ahead of time. It is the expected mass at the same time in the trajectory, if the vehicle has been running at minimum thrust for the entire flight.

After the relaxation and the variable change, the optimization problem becomes problem P3 found in Eqs. (43–53). The boxes highlight the differences between problems P2 and P3.

$$\min \boxed{-q(t_f)} \tag{43}$$

subject to

$$\dot{\boldsymbol{r}} = \boldsymbol{v} \tag{44}$$

$$\dot{\boldsymbol{v}} = \boxed{\boldsymbol{a}_t} - 2\boldsymbol{\omega}\times\boldsymbol{v} - \dot{\boldsymbol{\omega}}\times\boldsymbol{r} - \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}) + \nabla U(\boldsymbol{r}) \tag{45}$$

$$\boxed{\dot{q} = -\frac{1}{v_{\text{ex}}} a_{\text{tm}}} \tag{46}$$

$$\|\boldsymbol{a}_{\text{t}}\| \le a_{\text{tm}} \tag{47}$$

$$\boxed{T_{\min} \mathrm{e}^{-q_o}\left[1-(q-q_o)+0.5\,(q-q_o)^2\right] \le a_{\text{tm}}} \tag{48}$$

$$\boxed{a_{\text{tm}} \le T_{\max}\,\mathrm{e}^{-q_o}\,[1-(q-q_o)]} \tag{49}$$

$$\|\boldsymbol{r}-\boldsymbol{r}_f\|\cos\theta - (\boldsymbol{r}-\boldsymbol{r}_f)^T\hat{n} \le 0 \tag{50}$$

$$\boxed{q \ge q_{\text{dry}}} \tag{51}$$

$$\boldsymbol{r}(0) = \boldsymbol{r}_0,\ \boldsymbol{v}(0) = \boldsymbol{v}_0,\ \boxed{q(0) = \ln(m_{\text{wet}})} \tag{52}$$

$$\boldsymbol{r}(t_f) = \boldsymbol{r}_f, \qquad \boldsymbol{v}(t_f) = \boldsymbol{v}_f, \qquad t_f \text{ given} \tag{53}$$

Problem P3 is equivalent to problem P2 as long as the Taylor series expansion is a close approximation. The optimal point for problem P3 is the optimal solution of problem P2 and thus problem P1. With the exception of the gravitational acceleration, the dynamics are linear in state and control and the constraints are linear or second-order cones, yielding a convex optimization problem. The gravitational acceleration term will be handled via the successive solution method.

## IV. Successive Solution Method

The trajectory dynamics equations, Eqs. (44–46), in problem P3 are nonlinear, largely due to the nonlinear and complex gravitational acceleration terms that are functions of the spacecraft position vector. To use convex optimization, all equality constraints (including the state equations) must be linear. To this end, Eqs. (44–46) may be rearranged in the form of Eq. (54).

$$\dot{\boldsymbol{x}} = A(r)\boldsymbol{x} + B\boldsymbol{u} + c(r) \tag{54}$$

The full state is represented by $\boldsymbol{x} = [\boldsymbol{r}^T, \boldsymbol{v}^T, q]^T$ and the control vector by $\boldsymbol{u} = [\boldsymbol{a}_t^T, a_{\text{tm}}]^T$. All the nonlinear terms, that is, the gravitational acceleration, are located in the matrix $A$ and vector $c$. The matrix $A$ and vector $c$ are functions of only the vehicle position vector.

The successive solution approach first proposed in [17] is adopted and extended in this work to repeatedly approximate the nonlinear dynamics in Eq. (54) with linear dynamics. Let $r^{(k-1)}$ be the position vector of the vehicle in the $(k-1)$th successive solution (which is a function of time). Instead of Eq. (54), the dynamics used to find the $(k)$th solution are functions of the previous iteration's position vector, as shown in Eq. (55).

$$\dot{\boldsymbol{x}}^{(k)} = A(r^{(k-1)})\boldsymbol{x}^{(k)} + B\boldsymbol{u}^{(k)} + c(r^{(k-1)}) \tag{55}$$

The simplest choice for the first solution is $r^{(0)} = r_0$ (where $r_0$ is the initial condition of the radius vector). Now with $r^{(k-1)}$ known as an explicit function of time, the dynamics in Eq. (55) are linear and time-varying (for $k > 1$). Each successive problem can now be formulated and solved as a second-order cone program. Note that the linear dynamics in Eq. (55) are not obtained by the traditional linearization method based on first-order Taylor series expansions. When the successive solution sequence $\{\boldsymbol{x}^{(k)}\}$ converges, the dynamics in Eq. (55) are exactly the same as the original dynamics in Eq. (54), with no approximations involved. The tolerance used in this investigation to determine convergence is 0.5 m. A tighter tolerance can be achieved at the cost of an increase in number of iterations.

Extensive analysis examined the placement of the gravitational terms. The best placement in terms of the number of iterations required for convergence and the ease of changing gravity models is to place the dominant term in $A$ and the remaining higher-order terms in $c$ [see Eq. (56)]. The dominant term is defined in Eq. (57).

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \omega^2 + \text{dom} & 0 & 0 & 0 & 2\omega & 0 & 0 \\ 0 & \omega^2 + \text{dom} & 0 & -2\omega & 0 & 0 & 0 \\ 0 & 0 & \text{dom} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\frac{1}{v_{\text{ex}}} \end{bmatrix} \qquad c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{\partial U}{\partial r_x} - \text{dom} \\ \frac{\partial U}{\partial r_y} - \text{dom} \\ \frac{\partial U}{\partial r_z} - \text{dom} \\ 0 \end{bmatrix} \tag{56}$$

$$\text{dom} = \begin{cases} -\dfrac{\mu}{r^3} & 2\times 2,\ 4\times 4 \text{ model} \\ \alpha_{0,0} j_1\left(\dfrac{\alpha_{0,0} r}{R_b}\right)\bar{A}_{0,0,0} + \alpha_{1,0} j_1\left(\dfrac{\alpha_{1,0} r}{R_b}\right)\bar{A}_{1,0,0} + \alpha_{2,0} j_1\left(\dfrac{\alpha_{2,0} r}{R_b}\right)\bar{A}_{2,0,0} & \text{Bessel model} \end{cases} \tag{57}$$

For the $2\times 2$ and $4\times 4$, the dominant term, dom, is the degree and order of 0 term, which simplifies to the standard Newtonian gravity model. The dominant term for the Bessel model is the summation over the powers of degree and order 0, which contains three terms for the three powers. In Eq. (56), the rotation vector $\boldsymbol{\omega}$ in the equations of motion is assumed to be along the asteroid $+Z$ axis at a constant rate of $\omega$. Other rotation vectors can be accommodated by placing the corresponding terms in $A$.

The equations of motion, while now linear, are in continuous time; however, optimizers require the problem to be in discrete time points. To form a discrete system the problem is divided into $n$ number of time points with a fixed $dt$ time step between each point. Trapezoidal rule is used to propagate the trajectory between the points and is arranged so that each point is a function of the previous point and the coefficient matrices [see Eq. (58)].

$$\boldsymbol{x}_j = \left[I - 0.5dtA_j\right]^{-1}\Big[(I + 0.5dtA_{j-1})\boldsymbol{x}_{j-1} + 0.5dt\Big(B_j\boldsymbol{u}_j + B_{j-1}\boldsymbol{u}_{j-1} + c + c_{j-1}\Big)\Big] \tag{58}$$

Each of the $n$ points is represented this way, with $j$ being the current point and $j-1$ the previous point. The first point $n = 1$ is the initial condition of the vehicle and thus known. This form replaces the continuous dynamics in problem P3 as $n-1$ additional equality constraints levied on the problem.

In the analysis, a time step of 2.0 s was used. This was compared to smaller time steps of 1.0 and 0.5 s, with negligible change in propellant used and number of required iterations. The smaller time steps increase the number of equality constraints levied on the problem, thus increasing the number of calculations and simulation runtime. Two seconds yields good accuracy, while containing a fourth of the number of equality constraints as the 0.5 s case. Care should be taken to balance out the propagation accuracy and the number of equality constraints. A smaller time step may be necessary for asteroids with large irregular shapes, in order to better capture the changes in gravitational acceleration. If the flight time of the trajectory is not exactly divisible by the fixed time step, the time step will be adjusted slightly to allow each time step to be identical.

As a final step, the algorithm variables were scaled to keep the values near one and to keep any variable from having extra emphasis or weighting due to its magnitude. The optimization solver is sensitive to scaling, so care must be taken when choosing the scale factors. Scaling is important to ensuring good and consistent convergence in all cases.

All distance parameters are scaled by $R_{\text{sc}}$, velocity by $v_{\text{sc}}$, acceleration by $g_{\text{sc}}$, time by $t_{\text{sc}}$, angular rate by $\omega_{\text{sc}}$, and thrust by $T_{\text{sc}}$, as defined in Eq. (59). Variable names have not been changed to indicate scaled or not scaled. The scaled version of the variable is used in the algorithm calculations when coded.

$$R_{\text{sc}} = \gamma, \quad v_{\text{sc}} = \sqrt{R_{\text{sc}} g_{\text{sc}}}, \quad g_{\text{sc}} = \frac{\mu}{R_{\text{sc}}^2}, \quad t_{\text{sc}} = \sqrt{\frac{R_{\text{sc}}}{g_{\text{sc}}}}$$
$$\omega_{\text{sc}} = \sqrt{\frac{g_{\text{sc}}}{R_{\text{sc}}}}, \quad m_{\text{sc}} = 1, \quad \mu_{\text{sc}} = \mu, \quad T_{\text{sc}} = g_{\text{sc}} m_{\text{sc}} \tag{59}$$

The distance scale factor is set to the smallest semimajor axis value ($\gamma$), or an approximation of the shortest axis for irregularly shaped asteroids. This choice is consequential, because it keeps the scaled radius to no smaller than unity, which in turn will prevent the position terms, such as $1/r^3$, in the gravitational acceleration from becoming very large values. When the largest semimajor axis was used, the iterations failed to converge on a single answer. The mass is not scaled, as the natural log function acts as scaling function and keeps $q$ near one.

## V. Optimal Flight Time Determination

The propellant optimal convex optimization problem discussed in the previous sections assumed that a fixed flight time was given to the algorithm. Ultimately, the algorithm should determine the optimal flight time that yields minimum propellant usage over all flight times ($t_f^*$). Parametric sweeps of flight time showed that propellant usage is unimodal with respect to flight time (see Sec. VII for proof), which means that a simple single-variable optimization method will easily find the optimal flight time. This is achieved by wrapping the convex optimization problem with an outer optimization loop that minimizes propellant with respect to flight time. Brent's method was selected as the solver routine, as it only requires function evaluations of the convex optimization loop. It does not require derivatives or an approximation of the derivatives through differencing, which is advantageous given the complexity of the inner loop's optimization problem. For classes of functions that are twice differentiable near the minimum, of which the flight time is, Brent's method achieves superlinear convergence [28].

The goal is to run this algorithm onboard the spacecraft; therefore, computational time is a driving concern. Even though Brent's method is near superlinear convergence, the convex optimization trajectory design algorithm will need to be called 6–10 times. To decrease computational time, the discretization time step in the convex optimization algorithm was increased in order to determine the optimal flight time. Then, the inner loop trajectory design was re-evaluated for the optimal flight time using the original smaller discretization time step, yielding the final designed trajectory. A flow diagram for this process is located in Fig. 2.

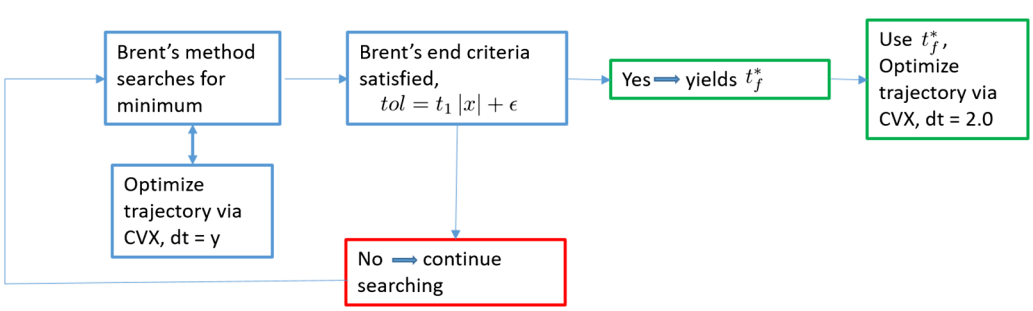


**Fig. 2 Flow diagram for the outer loop interactions with the inner loop.**

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**Table 1 Time step comparisons for Brent's method**

| | North Pole | | | Equatorial | | | Castalia LS3 | | |
|---|---|---|---|---|---|---|---|---|---|
| Time step, s | $t_f^*$, s | Prop used, kg | Run time, min | $t_f^*$, s | Prop used, kg | Run time, min | $t_f^*$, s | Prop used, kg | Run time, min |
| 2 | 487.69 | 4.9228 | 21.3 | 512.08 | 5.2868 | 16.5 | 512.6 | 5.340 | 7.2 |
| 5 | 487.78 | 4.9228 | 13.4 | 512.29 | 5.2868 | 8.2 | 517.9 | 5.340 | 3.7 |
| 10 | 488.03 | 4.9228 | 7.0 | 512.87 | 5.2868 | 5.0 | 513.4 | 5.340 | 2.2 |
| 15 | 489.55 | 4.9228 | 6.3 | 514.10 | 5.2868 | 4.2 | 523.0 | 5.341 | 2.0 |

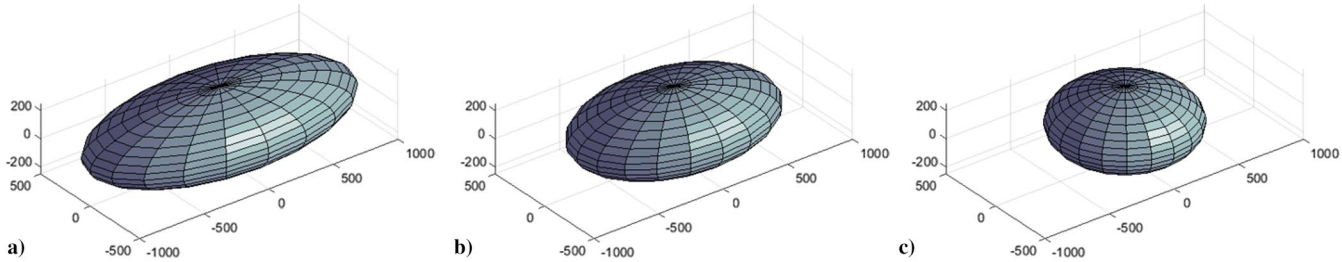


**Fig. 3 Three asteroids under investigation. a) 1000 × 500 × 250 m; b) 750 × 500 × 250 m; c) 500 × 500 × 250 m.**

Four discretization time steps were examined: 2, 5, 10, and 15 s. The choice of the best time step is a balance of computational time and required accuracy. The data examining these time steps for two different landing sites, a North Pole and an Equatorial, on a generic triaxial asteroid (1000 × 500 × 250 m) are located in Table 1. Table 1 also includes the same analysis for the third landing site on the asteroid Castalia. All three landing sites required seven function evaluations of the inner optimization loop. As the time step increases from 2 to 10 s, the required run time drops significantly to one third of the 2 s run time, with a change in optimal flight time of less than 1 s and no difference in propellant usage. The increase to 15 s does not decrease the run time significantly, while the difference in optimal flight time increases. A time step of 10 s for the initial calls to the inner optimization loop is the best balance between accuracy and computational speed.

## VI. Simulation Setup

The algorithm formulated above was rigorously analyzed. A testbed was programmed in Matlab using the publicly available convex optimization package CVX.[‡] CVX is designed to handle a wide range of convex optimization problems (see [29]). A tolerance of 0.5 m was used in the successive solution method to verify that the final trajectory and the trajectory immediately preceding it are close enough for the gravity model representation. Smaller tolerances are achievable at the cost of more iterations.

This is a three–degree-of-freedom analysis, so only the translational motion is investigated. The vehicle is represented as a point mass, without physical dimension. Only the basic vehicle characteristics, mass, thrust, and Isp, are needed for the analysis. The vehicle's mass is 1400 kg with 400 kg devoted to propellant. The 1000 kg of dry mass is devoted to vehicle structure and hardware along with payload. The engines have an Isp of 225 s and range in thrust with a maximum of 80 N and a minimum of 20 N. While this is considered low thrust, additional analysis was completed with a quarter of the original thrust, a maximum of 20 N, and a minimum of 5 N. These two thrust levels are sufficiently small to make the problem challenging, along with being representative of typical mission scenarios.

To show robustness, a variety of asteroid sizes, asteroid rotation speeds, and landing sites were examined. Three homogenous triaxial ellipsoidal asteroid sizes were examined: 1000 × 500 × 250 m (A1), 750 × 500 × 250 m (A2), and 500 × 500 × 250 m (A3). The three asteroids are depicted in Fig. 3. The gravitational coefficients for a triaxial ellipsoidal asteroid can be calculated analytically (see [22]). The asteroids rotate at a constant spin rate along the $+Z$ axis. The amount of time that it takes for the asteroid to complete one revolution is the rotation period. A period of 12, 8, 4, and 2 h were examined. These speeds are typical for asteroids.

**Table 2 Initial conditions for the trajectories**

| | $r_{0_x}$, m | $r_{0_y}$, m | $r_{0_z}$, m | $v_{0_x}$, m/s | $v_{0_y}$, m/s | $v_{0_z}$, m/s |
|---|---|---|---|---|---|---|
| North Pole | −50 | −50 | $1000+\gamma$ | 2 | 1 | 0 |
| Equatorial | $1000+\alpha$ | −50 | −50 | 0 | 2 | 1 |

$\gamma$ is the semimajor axis length corresponding to the $Z$ axis. $\alpha$ is the semimajor axis length corresponding to the $X$ axis.

Two landing sites were chosen for the triaxial ellipsoidal asteroid to bound landing on the asteroid. The North Pole trajectory lands on the $+Z$ axis. The pole of the asteroid is where the gravitational acceleration is the strongest, including the largest change of gravitational acceleration over the trajectory. The North Pole landing site is on the spin axis, so rotational effects are nonexistent. The second landing site is located on the $+X$ axis, corresponding to an Equatorial trajectory. This axis experiences the largest forces from rotational effects, as it is the longest semimajor axis. Also, the landing site does not stay at a fixed point in space and the trajectory end point must match the landing site's rotational speed. The trajectories cover 1000 m in altitude from their initial starting conditions until reaching the landing sites. As the asteroid size changes, the starting conditions are based off the semimajor axes lengths. The initial conditions located in Table 2 include uprange and out-of-plane position and velocity components that will need to be eliminated. The analysis was repeated using a second set of initial conditions, where the vehicle starts at a point directly over the landing site to indicate that it had hovered above the site before the descent (see [19]).

To demonstrate with a real and irregularly shaped asteroid, the asteroid 4769 Castalia was examined using the higher-fidelity gravity model, 4 × 4 combined with spherical Bessel. Castalia is an elongated binary asteroid that has been under investigation over the last 2 decades and has high-fidelity shape models, resulting in determination of gravitational parameters. A mesh model[§] shows the asteroid shape along with the Cartesian coordinate system used in this research (Fig. 4) [30]. To keep the orientation of the asteroid similar to subsequent plots, the $-Y$ axis is shown with the $+X$ and $+Z$ axes in the diagram. The rotation period is 4.07 h along the $+Z$ axis. Castalia is one of the asteroids used as a test of the spherical Bessel gravity model and has available coefficients for it [25] and the 4 × 4 spherical harmonics gravity model [31].

[‡]http://cvxr.com/cvx/download/ v2.0 [released 12 September 2013].

[§]http://sbn.psi.edu/pds/resource/rshape.html [accessed 17 March 2016].

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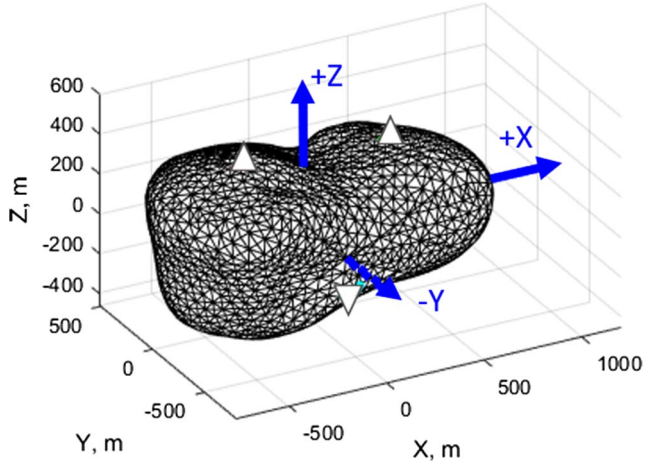


**Fig. 4 Asteroid Castalia showing the landing sites and the associated coordinate system axes. Note: $-Y$ is shown as opposed to $+Y$.**

**Table 3 Castalia landing site coordinates and associated velocity**

| | $r_{f_x}$, m | $r_{f_y}$, m | $r_{f_z}$, m | $v_{f_x}$, m/s | $v_{f_y}$, m/s | $v_{f_z}$, m/s |
|---|---|---|---|---|---|---|
| LS1 | −345 | −67 | 370 | 0.0287 | −0.148 | 0 |
| LS2 | 459 | 23.5 | 302 | −0.0101 | 0.197 | 0 |
| LS3 | 41.8 | −348 | −168 | 0.149 | 0.0179 | 0 |

**Table 4 Initial conditions for the Castalia trajectories**

| | $r_{0_x}$, m | $r_{0_y}$, m | $r_{0_z}$, m | $v_{0_x}$, m/s | $v_{0_y}$, m/s | $v_{0_z}$, m/s |
|---|---|---|---|---|---|---|
| LS1 | −1066.1 | −157.5 | 1060.6 | 1.097 | −1.823 | 0.689 |
| LS2 | 1323.3 | 18.2 | 809.8 | −0.652 | 1.969 | 0.835 |
| LS3 | 97.6 | −1227.8 | −645.3 | 2.037 | −0.190 | 0.902 |

Three landing sites on Castalia were analyzed and are referred to as LS1, LS2, and LS3. The landing sites move away from pure polar and equatorial sites due to the geometry of the asteroid and their locations. The coordinates of the landing sites are listed in Table 3 and marked on the asteroid in Fig. 4 [25]. The velocity of the landing site is the speed due to the rotation of the asteroid, listed in Table 3. The initial locations of the vehicle were chosen to mirror the initial starting points for the ellipsoidal asteroids. The trajectories all start at an altitude of 1000 m and include out of plane and uprange components that are eliminated in order to land softly. The values for the initial vehicle states are located in Table 4. The analysis was repeated using a second set of initial conditions, where the vehicle starts at a point directly over the landing site to indicate that it had hovered above the site before the descent (see [19]).

## VII. Testing

It was desired to show algorithm robustness by evaluating it with a range of asteroids, starting with homogenous triaxial ellipsoidal asteroids and then extending to the binary asteroid Castalia. The binary asteroid Castalia is a good example of an irregularly shaped asteroid. It is long and flat, which invalidates the triaxial asteroid assumptions and the use of the $2 \times 2$ gravitational model, especially near the surface. The results will focus on Castalia, as it uses the more complex gravity model and the triaxial ellipsoidal asteroid results mirror the Castalia results. Trends with respect to asteroid shape, size, and speed were completed with the triaxial ellipsoidal asteroids.

### A. Inner Optimization Loop

Parameter sweeps for the triaxial ellipsoidal asteroids were performed by varying the flight time, while designing the optimal trajectory for that flight time with the convex optimization method. Propellant usage as a function of flight time is shown in Figs. 5–7. The most important item to note is that propellant usage is unimodal with respect to flight time, the key trait that allows the outer loop optimization to be successful. Figure 5 compares the propellant usage for the different asteroid sizes. For the North Pole trajectories, the smaller the asteroid, the more propellant consumed during flight. This contrasts the Equatorial trajectory propellant consumption, which is fairly insensitive to asteroid size. Figure 6 compares propellant usage with respect to asteroid rotation period. The faster rotation speeds for the Equatorial trajectories require more propellant. The North Pole trajectory propellant consumption is insensitive to the rotation speed as expected. The quarter thrust trajectories showed similar trends. The Castalia flight time parameter sweep for all three landing sites is shown in Fig. 7. Propellant usage is similar between the three landing sites, despite the locations being on different parts of the asteroid.

The successive solution process typically took 2–5 iterations for the ellipsoidal asteroids and 3 iterations for Castalia to design the trajectory. The low number of required iterations shows stability in the successive solution method. Overall, the Equatorial landing site tended to use less iterations over the North Pole landing site. There were no noticeable trends with respect to asteroid size nor rotation speed for the number of iterations required.

By using the convex optimization coupled with the successive solution method, the thrust profile follows the bang-bang profile dictated by the Optimal Control Theory. This bang-bang refers to the thrust magnitude following the maximum or minimum thrust value and then rapidly changing to the other thrust bound. The trajectories fell into three categories: max-min-max, max-min, and minimum for the entire flight (all min). The shortest flight times followed the max-min-max and the longest flight times were all min.

Table 5 contains the flight times corresponding to the switches between the thrust profile categories for Castalia. The middle row corresponds to the start of the max-min thrust profiles and the bottom row corresponds to the switch to all min. The top line is the optimal flight time, smallest propellant usage. The optimal flight time occurs before the switch to max-min, though less than 10 s before the switch.

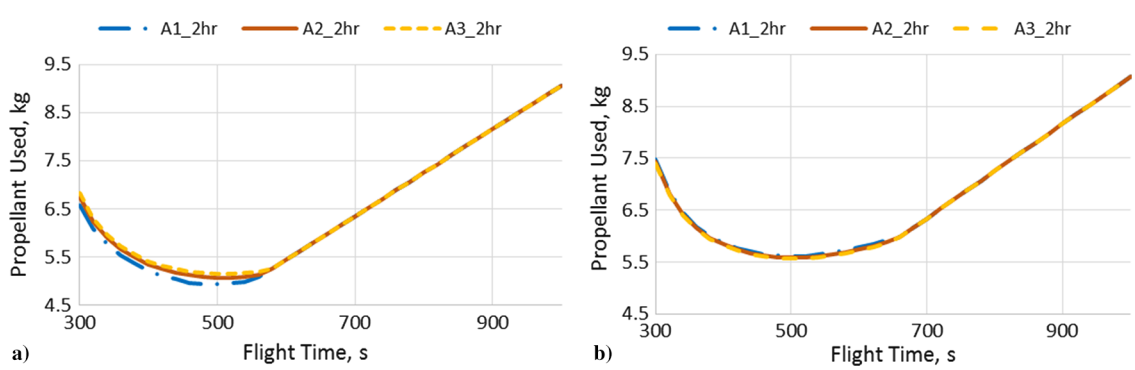


**Fig. 5 Propellant usage parameters sweeps for the different asteroid sizes with a 2 h period. a) North Pole; b) Equatorial.**

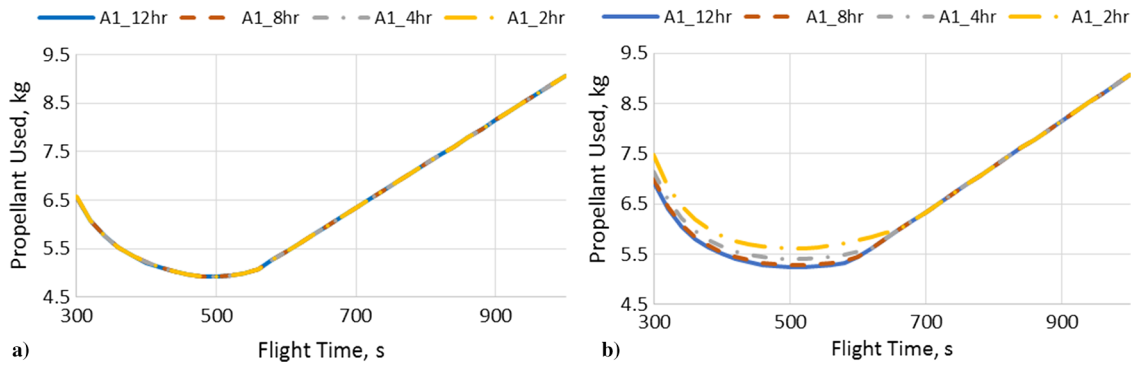


**Fig. 6 Propellant usage parameters sweeps for the 1000 × 500 × 250 m asteroid comparing the different rotation speeds. a) North Pole; b) Equatorial.**

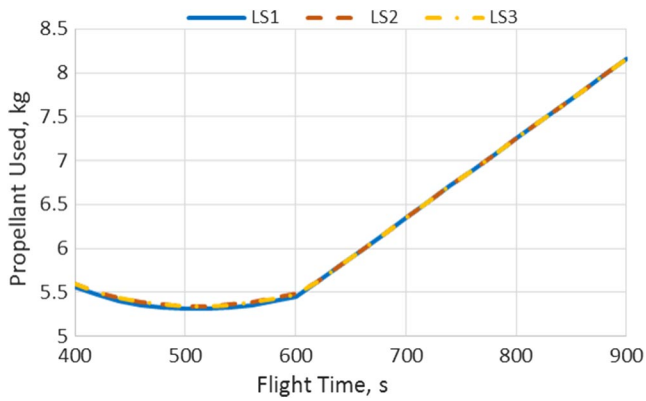


**Fig. 7 Propellant usage versus flight time for the three Castalia landing sites.**

**Table 5 Flight times corresponding to optimal propellant and thrust profile switches for the asteroid Castalia**

| | LS1 | LS2 | LS3 |
|---|---|---|---|
| Optimal prop | 513 s | 512 s | 513 s |
| Start max-min | 520 s | 521 s | 521 s |
| Start all min | 602 s | 608 s | 607 s |

**Table 6 Percent difference in mass from optimal propellant case for the asteroid Castalia**

| | LS1 | LS2 | LS3 |
|---|---|---|---|
| Start max-min | 0.02% | 0.02% | 0.01% |
| Start all min | 2.75% | 3.15% | 3.04% |

Propellant usage is flat in this regime, so the difference in optimal propellant usage and propellant usage at the switch to max-min is a small fraction of a percent (see Table 6). The propellant usage difference is within 3.5% between the all min and the optimal flight time. However, the propellant usage becomes steeper at this point. Therefore, it is advantageous to ensure that the mission flight time is before the switch to all min.

A 400 s flight time for LS3 was designed, in order to fully demonstrate a case that clearly designs the bang-bang thrust profile. The associated plots from this trajectory are located in Fig. 8. The difference between the acceleration slack variable and the magnitude of the acceleration vector is less than $6.0 \times 10^{-5}$%, demonstrating that the lossless convexification assumption is valid and the relaxed problem is the original problem for the optimal solution.

### B. Combined Outer and Inner Loop Optimization

As the ultimate goal is to design the optimal propellant trajectory over all flight times and trajectories, the outer loop optimization (Brent's method) wrapped around the inner optimization algorithm (CVX) was tested with the Castalia trajectories. For all sets of trajectories, the full optimization method was successful. Table 7 contains the optimal flight time and the corresponding propellant usage for that flight time. The last column is the number of times the inner loop designed a trajectory. Each of these inner loop executions contained multiple iterations for the successive solution method. The quarter thrust required more calls to the trajectory design loop than the full thrust.

The propellant optimal designed trajectories corresponding to the optimal flight time for LS1, LS2, and LS3 are shown in Fig. 9. Figure 10 compares the magnitude of the velocity relative to the landing site velocity, the thrust magnitude profile, a check on the slack variable, and the mass profiles for these three trajectories. All three trajectories have the max-min-max thrust profile, though the final maximum thrust is only for a few seconds. The difference between the slack variable and the magnitude of the acceleration vector is less than $1 \times 10^{-4}$%. The trajectory parameters for the

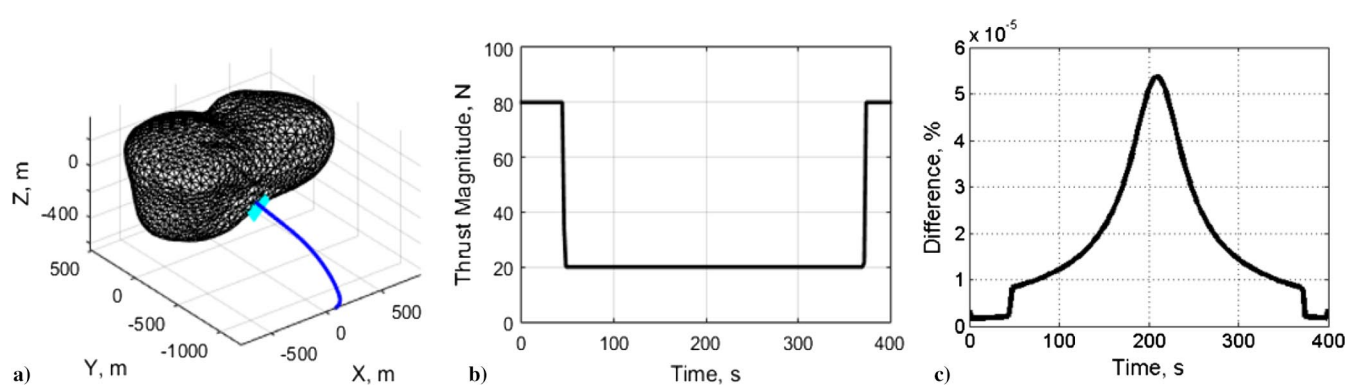


**Fig. 8 LS3 trajectory for a flight time of 400 s. a) 3-D vehicle position; b) thrust magnitude; c) difference between the slack variable and the acceleration vector magnitude.**

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**Table 7 Comparison of the optimal flight time, propellant used, and number of inner loop executions for the Castalia trajectory configurations**

| | Optimal flight time, s | Optimal prop used, kg | Number of inner loop executions |
|---|---|---|---|
| LS1 | 512.86 | 5.31 | 7 |
| LS2 | 512.27 | 5.34 | 7 |
| LS3 | 513.35 | 5.34 | 7 |
| LS1 quarter thrust | 1052.0 | 3.39 | 10 |
| LS2 quarter thrust | 1050.6 | 3.45 | 11 |
| LS3 quarter thrust | 1075.6 | 3.44 | 9 |

quarter thrust case are shown in Fig. 11, where the thrust magnitude and velocity magnitude profiles show more variation between the three landing sites as compared with the other cases.

## VIII. Additional Trajectory Constraints

Following [15,16], two additional trajectory constraints were also considered in the propellant optimal trajectory design problem and analyzed. The first constraint keeps the vehicle directly above the landing site with no lateral motion for the last few seconds of flight, or only allows vertical motion. The second constraint is the glide slope constraint. The original problem included a glide slope constraint; however, it was only used to ensure that the vehicle did not fly subsurface. Figure 12 depicts the glide slope constraint. It also illustrates the solely vertical motion via the spacecraft at the bottom of the depiction.

### A. Solely Vertical Motion Near the Landing Site Constraint

As the vehicle nears the landing site it is desirable and advantageous for it to be directly over the landing site with the only motion being a change in altitude. This allows the vehicle to land upright and removes tip over concerns. A new constraint was applied that eliminated the lateral motion during the last seconds of flight, resulting in solely an altitude change.

The new constraint is applied as two new equality constraints. First, the position vector from the landing site to the spacecraft is determined, Eq. (60).

$$\boldsymbol{r}_{ls} = \boldsymbol{r} - \boldsymbol{r}_f \tag{60}$$

This position vector, relative to the landing site, is used in the constraint instead of the position vector from the asteroid center to the

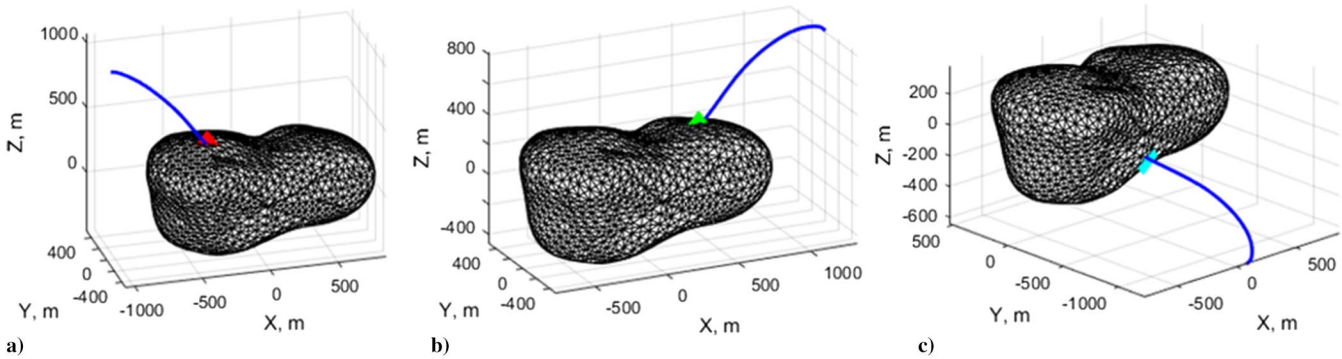


**Fig. 9 Optimal flight time trajectories landing on Castalia for a) LS1, b) LS2, and c) LS3.**

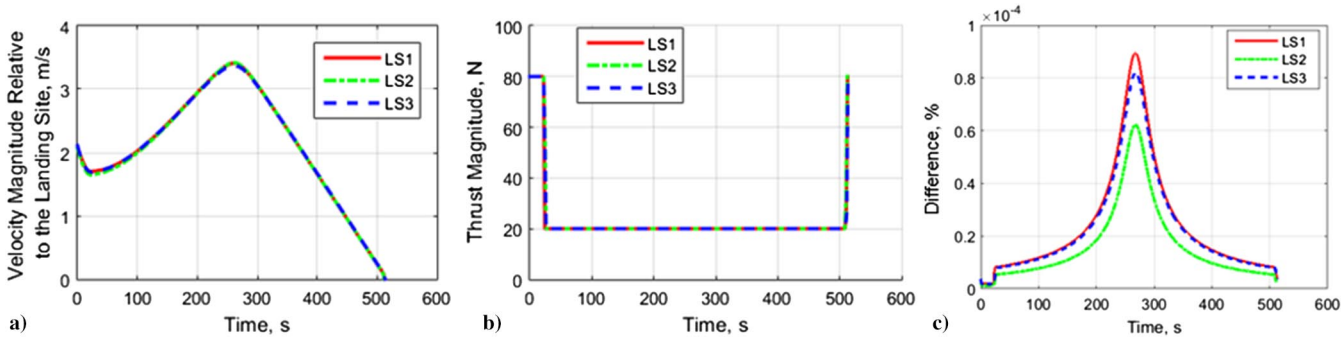


**Fig. 10 Optimal flight time trajectories landing on Castalia comparing the three landing sites. a) Velocity magnitude relative to the landing site; b) thrust magnitude; c) difference between the slack variable and the acceleration vector magnitude.**

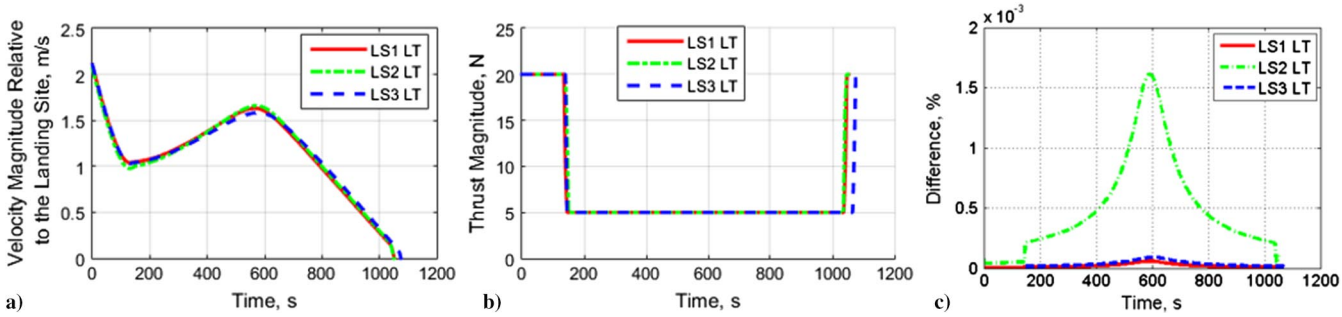


**Fig. 11 Optimal flight time trajectories landing on Castalia comparing the three landing sites for the quarter thrust cases. a) Velocity magnitude relative to the landing site; b) thrust magnitude; c) difference between the slack variable and the acceleration vector magnitude.**

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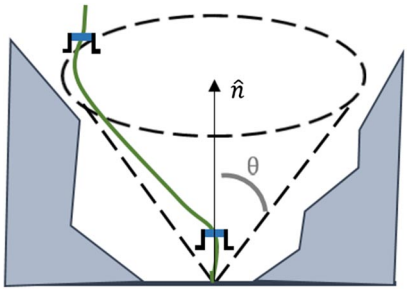


**Fig. 12 Glide slope trajectory constraint depiction.**

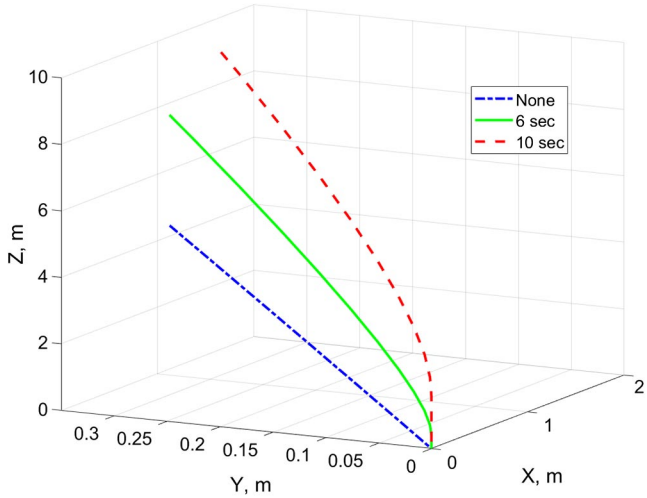


**Fig. 13 End of the North Pole trajectory with a 480 s flight time landing on a $1000 \times 500 \times 250$ m asteroid with an 8 h period without the vertical constraint, with a 6 s constraint, and with a 10 s constraint. Origin is the landing site.**

spacecraft as the vertical and horizontal vehicle motion is with respect to the landing site. The new equality constraints are located in Eq. (61).

$$\boldsymbol{r}_{ls}^T \hat{t}_1 = 0 \quad \boldsymbol{r}_{ls}^T \hat{t}_2 = 0 \tag{61}$$

The basis describing the plane tangent to the landing site is contained in $\hat{t}_1$ and $\hat{t}_2$. This can be any basis as long as the plane is completely described. A zero dot product with the tangent plane requires the motion to be solely normal to the landing site plane, which only allows motion in the vertical direction.

The North Pole trajectory with a 480 s flight time landing on the $1000 \times 500 \times 250$ m asteroid with an 8 h period was analyzed. The constraint was enforced during the last 6 s of flight and then during the last 10 s. The end portion of flight is shown in Fig. 13, comparing the trajectory without this constraint, and enforcing it for the last 6 and 10 s. The trajectory shape within a 2 m altitude band above the landing site differs greatly between the three trajectories. The propellant usage changed very slightly between the three cases, with the original case using 4.92 kg, the 6 s using 4.93 kg, and the 10 s using 4.94 kg. Addition of this constraint did not adversely affect the successive solution convergence. The difference between the slack variable and the acceleration vector was on the order of $4 \times 10^{-5}\%$. This shows that the optimal solution of the relaxed problem is still the optimal solution of the original problem.

Problems arose when enforcing the solely vertical motion near the landing site constraint for the Equatorial and Castalia trajectories. The low thrust of the vehicle (80 N) is not strong enough for the vehicle to match the landing site's velocity, while removing the lateral velocity. The velocity required to remain above the landing site changes with altitude. Originally, it was thought that the centripetal acceleration was the driver behind the vehicle being unable to eliminate the horizontal motion near the landing site; however, this is not the case [19].

Increasing the rotational period, which decreases both the landing site velocity and the centripetal and Coriolis accelerations, allows the problem to be solved. For the $1000 \times 500 \times 250$ m Equatorial case, the rotational period had to be increased to 92 h to enforce the vertical constraint for the last 6 s of flight. An alternative to show that the vehicle thrust is insufficient for the landing site velocity is to increase the thrust of the vehicle until it successfully lands. The upper thrust bound must be increased to 920 N as compared with 80 N for the $1000 \times 500 \times 250$ m asteroid with an 8 h period. A landing site with a moderate velocity, such as the Equatorial or the Castalia landing sites, is not recommended if the vehicle needs to completely eliminate lateral motion near the landing site with a low-thrust vehicle.

### B. Glide Slope Constraint

The glide slope constraint requires the vehicle to stay inside a cone centered at the landing site as shown in Fig. 12. The cone is described by the angle $\theta$ measured from the vector normal to the landing site, $\hat{n}$. The constraint is applied in the second-order cone form given in Eq. (62).

$$\|\boldsymbol{r} - \boldsymbol{r}_f\| \cos\theta - (\boldsymbol{r} - \boldsymbol{r}_f)^T \hat{n} \leq 0 \tag{62}$$

This equation is part of the original problem formulation; however, it was only used to prevent the vehicle from flying subsurface, corresponding to a $\theta$ of 90 deg.

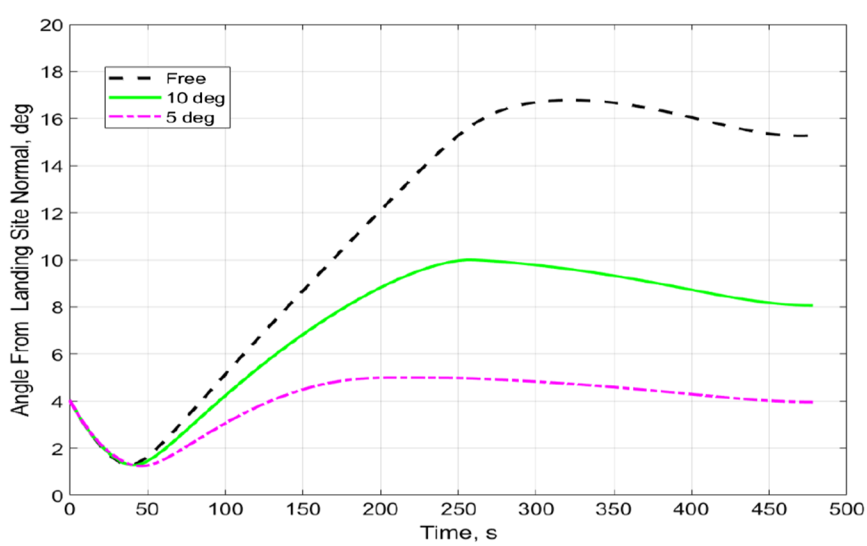


**Fig. 14 Angle from the landing site normal for the North Pole trajectory on $1000 \times 500 \times 250$ m asteroid with a 480 s flight time.**

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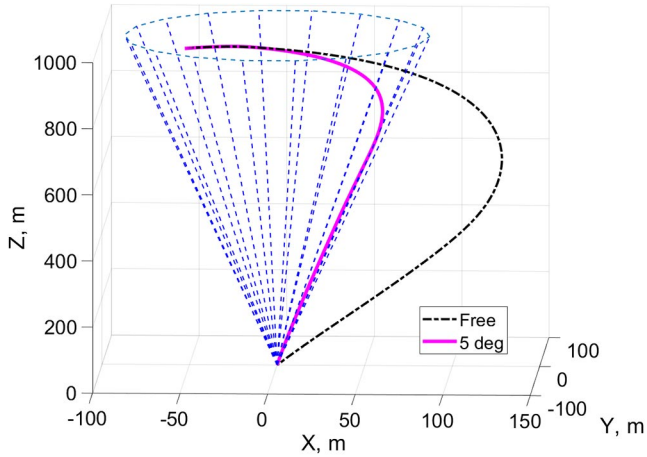


**Fig. 15 Comparison of trajectory without glide slope constraint (free) and 5 deg cone glide slope constraint relative to the landing site for the North Pole trajectory on a $1000 \times 500 \times 250$ m asteroid with a 480 s flight time.**

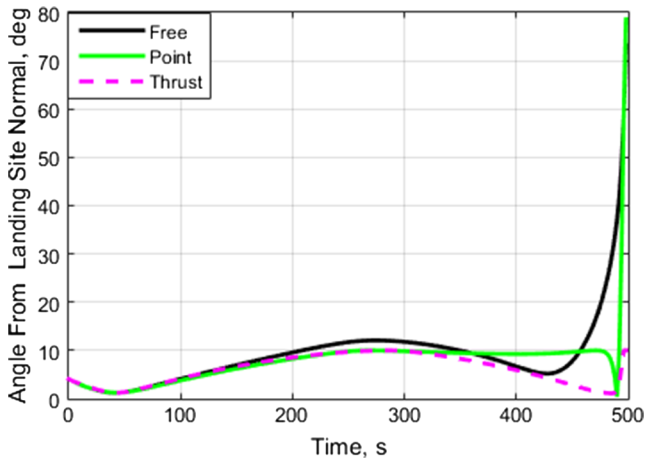


**Fig. 16 LS2: Angle from landing site normal with a 500 s flight time.**

As with the solely vertical motion near the landing site constraint, the glide slope constraint worked well for the North Pole trajectories, while issues existed for Equatorial and Castalia. A North Pole trajectory with a 480 s flight time landing on the $1000 \times 500 \times 250$ m asteroid with an 8 h period was run without the glide slope constraint (free, 90 deg), then with a 10 deg cone around the landing site, and finally with a 5 deg cone. The plot in Fig. 14 compares the angle between the landing site normal vector and the vehicle position vector. The free trajectory had an angle greater than 15 deg for the last 230 s of flight. When the 10 deg cone constraint is included, the entire trajectory takes a different shape with the 10 deg limit only being active for a single point. For the 5 deg cone constraint, the trajectory hovers near the 5 deg limit. The 3-D plot comparing the original trajectory to the 5 deg trajectory is located in Fig. 15. The dashed cone represents a 5 deg cone around the landing site.

The relaxation proof did not include the glide slope constraint. The difference between the acceleration slack variable and the magnitude of the acceleration vector was less than $6.0 \times 10^{-3}\%$. Thus, the relaxation holds for these cases and it is reasonable to assume that it will hold for other cases. Similar proofs including pointing constraints, which are similar to the glide slope constraint, exist and give confidence in lossless convexification with the glide slope constraint [16].

Near the landing site the cone around the landing site narrows and approaches the solely vertical motion near the landing site constraint. Given this similarity, it is not surprising that the Equatorial and Castalia cases encounter difficulties. To show that it is feasible to include the glide slope constraint and that this is a physical limitation of the vehicle and not a failure of method, the Castalia trajectories were run two different ways with the glide slope constraint enabled. The first run does not apply the constraint for the last 6 s of flight. This allows the vehicle to fly outside the cone near the landing site where the constraint is extremely tight. The second run increased the vehicle thrust and found that the thrust must be at least 320 N in order to fly a 10 deg cone with the glide slope constraint applied.

A 500 s flight time trajectory was designed for these two options, along with a trajectory without the glide slope constraint enabled. In the legend, "Free" refers to the run without the constraint applied, "Point" refers to the run that did not apply the constraint near the landing site, and "Thrust" refers to the 320 N case. Figure 16 compares the angle between the vehicle position vector and the landing site normal for LS2. The 3-D plot of the trajectory and the 10 deg cone around the landing site is located in Fig. 17, with the zero point corresponding to the landing site. The left plot is the entire trajectory, whereas the right zooms in on the landing site. The vehicle path stays near the cone boundary for most of flight for both of the constrained trajectories, whereas the free trajectory exceeds the cone. While it can be seen that the free trajectory deviates from the cone, the trajectory that does not constrain the last points leaves the cone very slightly, so slight that it is not visible. Therefore, it is feasible to refrain from constraining the last points and still get a trajectory that does not significantly leave the cone around the landing site.

## IX. Conclusions

This paper has shown that a propellant optimal trajectory for a soft landing on an irregularly shaped asteroid can be rapidly and reliably designed for a fixed flight time using convex optimization. In contrast to the soft landing problem on a planet with near-constant gravity, the gravitational potential model based on the standard Legendre polynomial expansion of spherical harmonics may not even be applicable for an irregularly shaped asteroid at low altitudes. Therefore, a major challenge is the nonlinearity in the propellant optimal asteroid landing problem. This nonlinear optimal control problem was cast into a relaxed problem through relaxation

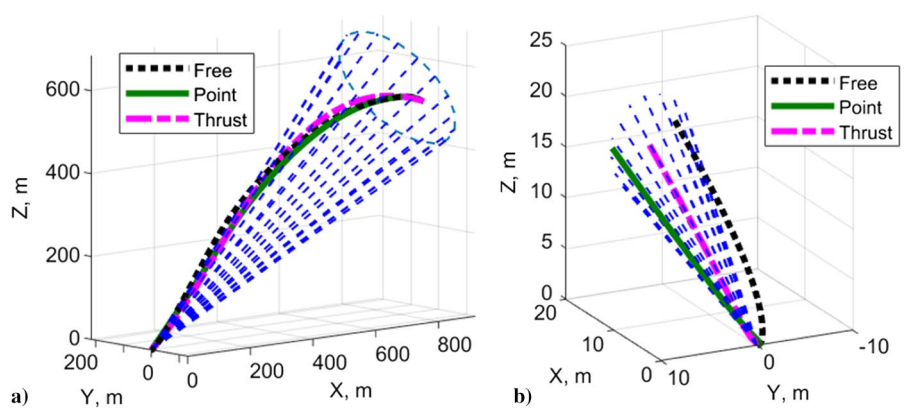


**Fig. 17 Ten-degree cone from LS2 with a 500 s flight time. a) Entire trajectory; b) focused on the landing site.**

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techniques. The optimal solution of the relaxed problem was theoretically proven to be the optimal solution of the original problem. The relaxed problem was then solved by a successive solution method as a sequence of second-order cone programs with no user-supplied initial guesses. Because optimal propellant usage is shown to be unimodal with respect to flight time, an outer loop using Brent's method was employed to determine the overall optimal flight time. A key finding is that the method examined in this paper proves to be applicable for increased levels of gravity model fidelity, as long as the gravitational model depends only on the spacecraft's position vector. Convergence is always achieved with a small number of successive solutions, whenever a solution exists. There is a balance between accuracy and computational speed in the methodology, which can be achieved through the tuning of step sizes and successive solution tolerances. Findings also include the importance of properly scaling the equations and the placement of the gravitational terms for good methodology robustness. Through extensive numerical applications, the proposed method is demonstrated to be a powerful tool for rapid optimal trajectory design and feasibility studies subject to various realistic and critical constraints. The high reliability, rapid convergence, and fully automated operation of this approach are appealing characteristics for potential autonomous onboard applications.

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